

Effect of Low-fat Diets on Plasma Levels of NFκB-regulated Inflammatory Cytokines and Angiogenic Factors in Men with Prostate Cancer

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Diet, nutritional status, and certain dietary supplements are postulated to influence the development and progression of prostate cancer. Angiogenesis and inflammation are central to tumor growth and progression, but the effect of diet on these processes remains uncertain. We explored changes in 50 plasma cytokines and angiogenic factors (CAFs) in 145 men with prostate cancer enrolled in a pre-operative, randomized controlled phase-II trial with four arms: control (usual diet); low-fat (LF) diet; flaxseed-supplemented (FS) diet; and flaxseed-supplemented, low-fat diet. The mean duration of dietary intervention was 30–31 days. Among the individual arms, the largest number of significant changes (baseline vs pre-operative follow-up) was observed in the LF arm, with 19 CAFs decreasing and one increasing ($p < .05$). Compared to the control arm, 6 CAFs—including pro-angiogenic factors (stromal-cell derived-1 α and myeloid factors (granulocyte-colony-stimulating factor, macrophage colony-stimulating factor — all decreased in the LF arm compared to controls; 3 and 4 CAFs changed in the FS and FS+LF arms, respectively. Weight loss occurred in the LF arms and significantly correlated with VEGF decreases ($P < 0.001$). The CAFs that changed in the LF arm are all known to be regulated by nuclear factor-kappa B (NF- κ B), and a pathway analysis identified NF- κ B as the most likely regulatory network associated with these changes in the LF arm, but not in the FS-containing arms. These results suggest that a low-fat diet without flaxseed may reduce levels of specific inflammatory cytokines and angiogenic factors and suggests that the NF- κ B pathway may be a mediator of these changes.